# CHRISTINA DURÓN

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## ACADEMIC EMPLOYMENT

**Associate Professor of Mathematics** *Aug 2026 – Present*
*Natural Science Division of Seaver College, Pepperdine University*

**Assistant Professor of Mathematics** *Aug 2022 – July 2026*
*Natural Science Division of Seaver College, Pepperdine University*

**Postdoctoral Research Associate** *Aug 2019 – May 2022*
*Department of Mathematics, University of Arizona*

**High School Teacher** *Aug 2013 – June 2019*
*Mathematics Department, The Webb Schools of California*

## RESEARCH INTERESTS

Network Theory; Network Dynamics; Statistical Analysis and Modeling of Complex Networks; Mathematical-Biology

## EDUCATION

**Claremont Graduate University** *May 2019*
*Ph.D. in Mathematics*

- **Thesis**: The Distribution of Betweenness Centrality in Exponential Random Graph Models
- **Advisors**: Dr. Ami Radunskaya (Professor, Pomona College) and Dr. Johanna Hardin (Professor, Pomona College)

**University of Washington** *June 2013*
*Master in Science in Applied Mathematics*

**Swarthmore College** *May 2012*
*Bachelor of Arts in Mathematics; Minor in Computer Science*

## PUBLICATIONS *

14. Kravitz H, **Durón C**, Brio M. Epidemic Dynamics Under Pulse Population Exchange. *Dynamics*, 6(3), 29. doi: 10.3390/dynamics6030029

13. Kravitz H, **Durón C**, Nieves B, Brio M. (2025). Data-driven optimization and parameter estimation for a metric graph epidemic model with applications to COVID-19 spread in Poland: A real-world example of optimization for a challenging Rosenbrock-type objective function. *An International Journal of Optimization and Control: Theories & Applications* (IJOCTA), 15(4). doi: 10.36922/IJOCTA025220106

12. **Durón C**, Kravitz H, Brio M. (2025). Edge percolation centrality: A new measure to quantify the influence of edges during percolation in networks. *PLoS One* 20(9): e0331475. doi: 10.1371/journal.pone.0331475

11. **Durón C**, Kravitz H, Brio M. (2025). Kolmogorov–Smirnov-Based Edge Centrality Measure for Metric Graphs. *Dynamics*, 5(2), 16. doi: 10.3390/dynamics5020016

* Authors are ordered by contribution.
† Undergraduate student.

10. Pfeffer D, **Durón C**. (2025). An Unlikely Duo: Injecting Art Projects in the Mathematics Classroom. *Journal of Humanistic Mathematics*, 15(1), 87-108. doi: 10.5642/jhummath.RDDV3141

9. Kravitz H, **Durón C**, Brio M. (2024). A Coupled Spatial-Network Model: A Mathematical Framework for Applications in Epidemiology. *Bulletin of Mathematical Biology*, 86(132): 1 – 35. doi: 10.1007/s11538-024-01364-3

8. **Durón C**, Swansen B[†], Kravitz H. (2024). Difference Approximation. *Wiley StatsRef: Statistics Reference Online* (eds N. Balakrishnan, T. Colton, B. Everitt, W. Piegorsch, F. Ruggeri and J.L. Teugels). doi: 10.1002/9781118445112.stat08440

7. Sullivant N, **Durón C**, Pfeffer D. (2023). From Mirrors to Wallpapers: A Virtual Math Circle Module on Symmetry. *Journal of Math Circles*: Vol. 3: Iss. 1, Article 1. Available at `https://digitalcommons.cwu.edu/mathcirclesjournal/vol3/iss1/1`.

6. **Durón C**. (2022). Adaptive Quadrature. *Wiley StatsRef: Statistics Reference Online* (eds N. Balakrishnan, T. Colton, B. Everitt, W. Piegorsch, F. Ruggeri and J.L. Teugels). doi: 10.1002/9781118445112.stat08388

5. **Durón C**, Farrell A. (2022). A Mean-Field Approximation of SIR Epidemics on an Erdös-Rényi Network Model. *Bulletin of Mathematical Biology*, 84(7): 1 – 19. doi: 10.1007/s11538-022-01026-2

4. **Durón C**. (2021). Linear Algebra, Computational. *Wiley StatsRef: Statistics Reference Online* (eds N. Balakrishnan, T. Colton, B. Everitt, W. Piegorsch, F. Ruggeri and J.L. Teugels). doi: 10.1002/9781118445112.stat00459.pub2

3. **Durón C**. (2020). Heatmap Centrality: A New Measure to Identify Super-Spreader Nodes in Scale-Free Networks. *PLoS ONE*, 15(7): e0235690. doi: 10.1371/journal.pone.0235690

2. **Durón C**, Pan Y, Gutmann D.H., Hardin J, & Radunskaya A. (2018). Variability of Betweenness Centrality and Its Effect on Identifying Essential Genes. *Bulletin of Mathematical Biology*, 81(9): 3655 – 3673. doi: 10.1007/s11538-018-0526-z

1. Pan Y, **Durón C**, Bush E.C., et al. (2018). Graph Complexity Analysis Identifies an ETV5 Tumor-Specific Network in Human and Murine Low-Grade Glioma. *PLoS ONE*, 13(5): e0190001. doi: 10.1371/journal.pone.0190001

# IN PREPARATION [*]

3. **Durón C**, Kravitz H, Brio M, Parkinson C, Lucas B[†]. Epidemic Thresholds on Adaptive Networks with Noncompliance. *Mathematical Biosciences*.

2. O'Brien E, **Durón C**. The Wasserstein metric as a tool to assess burn-in and convergence. *Statistical Science*.

1. **Durón C**, Mello J[†], Kamps K[†]. Network Evaluation of Influential Sensors on Critical Infrastructures. *The Journal of Dam Safety*.

# RESEARCH POSITIONS

**Graduate Research Assistant** *Jan 2017 – June 2018*
*Pomona College*

- NIH (1R01-CA195692-01) funding under Dr. Ami Radunskaya and Dr. Johanna Hardin

**Jet Propulsion Laboratory (JPL) Intern** *June 2015 – August 2015*
*California Institute of Technology*

- Implemented the Extended Kalman Filter (EFK) and incorporated inter-robot measurements to improve the state estimation and localization of autonomous vehicles

**Mathematical and Theoretical Biology Institute Researcher** *June 2011 – July 2011*
*Arizona State University*

- Developed a mathematical model for the evaluation and analysis of the air pollution in Los Angeles

[*] Authors are ordered by contribution.
[†] Undergraduate student.

# TEACHING EXPERIENCE

**Instructor of Record**
*Pepperdine University*

- **MATH 450: Mathematical Statistics** — *Spring 2024*
- **MATH 350: Mathematical Probability** — *Fall 2025, Fall 2023*
- **MATH 345: Numerical Methods** — *Spring 2026*
- **MATH 299: Directed Studies** — *Summer 2026*
- **MATH 260: Linear Algebra** — *Fall 2026, Spring 2026, Fall 2024, Spring 2023*
- **MATH 150: Calculus I** — *Fall 2026, Fall 2025, Fall 2024, Spring 2024, Fall 2023, Spring 2023, Fall 2022*

**Instructor of Record**
*University of Arizona*

- **MATH 491: Undergraduate Teaching Assistantship (UTA) Seminar** — *Fall 2021 – Spring 2022*
- **MATH 196M: Calculus I Supplementary Seminar** — *Spring 2022*
- **MATH 396L: Wildcats Proofs Workshop** — *Spring 2022*
- **MATH 464: Theory of Probability** — *Fall 2021*
- **MATH 363: Introduction to Statistical Methods** — *Spring 2021*
- **MATH 129: Calculus II** — *Fall 2020*
- **MATH 475A: Mathematical Principles of Numerical Analysis** — *Fall 2020*
- **MATH 163: Basic Statistics** — *Spring 2020*
- **MATH 122B: First Semester Calculus** — *Fall 2019*
- **MATH 196L: Precalculus Supplementary Seminar** — *Fall 2019*

**Instructor of Record**
*The Webb Schools of California*

- **Advanced Placement Computer Science Principles** — *Fall 2018 – Spring 2019*
- **Honors Precalculus** — *Fall 2014 – Spring 2019*
- **Introduction to Computer Programming with Python** — *Fall 2014 – Spring 2018*
- **Precalculus** — *Fall 2013 – Spring 2019*
- **Integrated Mathematics 2** — *Fall 2013 – Spring 2014*

# PRESENTATIONS

**Student-Directed**

4. **Modeling Epidemics with Noncompliance: A Discrete-Time Mean-Field Approach** *April 2026*
*Pepperdine University; Seaver College Research and Scholarly Research Symposium*

3. **Network Evaluation of Influential Sensors: A Proposed Approach** *March 2024*
*Pepperdine University; Seaver College Research and Scholarly Research Symposium*

2. **Using Networks to Evaluate an Infrastructure's Condition During Seismic Activity: A Proposed Approach** *Jan 2024*
*Joint Mathematics Meeting: AMS - PME Undergraduate Student Poster Session*

1. **Using Networks to Evaluate an Infrastructure's Condition During Seismic Activity: A Proposed Approach** *July 2023*
*Pepperdine University, Natural Science Division Summer Research*

**Contributed**

9. **Edge Percolation Centrality: Measuring Dynamic Influence on Networks** *March 2026*
*American Mathematical Society Special Session on Metric Graphs: Theory and Applications*

8. **Digitally-Enabled, Evidence Based Homework: Developing a Custom Platform to Improve Calculus Learning** *January 2026*
*Joint Mathematics Meeting Special Session on Digitally-Enabled Evidence-Based Teaching Practices*

7. **A Coupled Spatial-Network Model for Epidemiology** *July 2024*
*Frankfurt School of Finance and Management Workshop on Metric Networks*

6. **Math Circles: Elevating Student Engagement and Mathematical Enjoyment Outside the Classroom** *Jan 2024*
*University of Arizona Mathematics Educator Appreciation Day Conference (MEAD)*

5. **Incorporating Undergraduate Students in Research on Assessing Dam Safety Using Network Analysis** *Jan 2024*
*Joint Mathematics Meeting Special Session on Navigating the Benefits and Challenges of Mentoring Students in Data-Driven Undergraduate*
*Research Projects*

4. **Tiling a Chessboard: A Problem Adapted for the Virtual Math Circle** *Jan 2023*
*Joint Mathematics Meeting Special Session on Math Circle Activities as a Gateway into Mathematics*

3. **A Mean Field Approximation of SIR Epidemics on an Erdös-Rényi Network Model** *May 2021*
*Los Alamos-Arizona Days Conference (Virtual Poster)*

2. **Identifying Super-Spreader Nodes in Scale-Free Networks using Network Centrality Measures** *Sept 2020*
*Arizona Postdoctoral Research Conference (Virtual Talk)*

1. **Identifying Treatment Targets for Pediatric Gliomas using Network Centrality Measures** *June 2020*
*SIAM Conference on the Life Sciences (Virtual Talk)*

**Seminar**

4. **A Coupled Spatial-Network Model for Epidemiology** *Nov 2023*
*University of Florida, Health Systems Medicine Seminar (Virtual Talk)*

3. **Network Centrality: Theory to Applications** *Oct 2021*
*Arizona State University, Mathematical Biology Seminar (Virtual Talk)*

2. **Heatmap Centrality: A New Measure to Identify Super-Spreader Nodes in Scale-Free Networks** *Feb 2021*
*Claremont Colleges and University of Utah, Joint Applied Mathematics Seminar (Virtual Talk)*

1. **Network Data Analysis Techniques on DESeq and RNASeq Data** *Nov 2019*
*University of Arizona, TRIPODS Research Working Group 6 - Analyzing large-scale point-set data*

### Industry

1. **Network Evaluation of Influential Sensors: A Proposed Approach**‡ *March 2024*
   *California Department of Water Resources*

2. **Network Evaluation of Influential Sensors: A Proposed Approach**‡ *Nov 2023*
   *Southern California Edison, Dam & Public Safety Group*

## PROFESSIONAL DEVELOPMENT

### Program

- **American Institute of Mathematics SQuaRE Program** *July 2026, April 2025*
  *California Institute of Technology*
  - Member of a multi-year research collaboration focused on creating computational frameworks to predict epidemic transitions and optimize social measures for disease mitigation.

- **Project NExT** (Green 2023) *Aug 2023 – Aug 2024*
  *Mathematical Association of America (MAA)*
  - Focused on all aspects of an academic career: improving the teaching and learning of mathematics, engaging in research and scholarship, finding exciting and interesting service opportunities, and participating in professional activities

### Certification

- **Diversity, Equity, and Inclusion in the Workplace** *May 2021*
  *University of South Florida*
  - Focused on ways that organizations can create a more diverse workplace, address equity issues, and foster inclusivity

- **Effective Online Discussions** *June 2020*
  *University of Arizona*
  - Developed strategies for designing and facilitating effective online discussions that deepen learning, expand student exposure to curriculum, and increase student engagement

- **Teaching the Large Online Course** *June 2020*
  *University of Arizona*
  - Developed instructional practices for encouraging student engagement and motivation in a large online class, as well as for effectively managing administrative tasks such as monitoring student progress and conducting assessments

### Workshops

- **Workshop on Metric Networks** *July 2024*
  *Frankfurt School of Finance and Management*

- **Broadening Participation: 2024 Mathematical and Physical Sciences Workshop for New Investigators** *July 2024*
  *National Science Foundation*

- **Network Modeling for Epidemics** *Aug 2020*
  *University of Washington*

- **BioBridge Clinic** *Jan 2020*
  *University of California, Irvine*

- **Computational Genomics Summer Institute** *May 2019*
  *University of California, Los Angeles*

‡ Presented with undergraduate students.
‡ Presented with undergraduate students.

# SERVICE

### Mentoring and Advising

- **Undergraduate Research Advisor** (Team of 1 undergraduate) *Summer 2026 – Present*
  *Pepperdine University*
- **Undergraduate Research Advisor** (Team of 3 undergraduates) *Fall 2025 – Spring 2026*
  *Pepperdine University*
- **Tidy Thursday Advisor** *Spring 2026, Fall 2024*
  *Pepperdine University*
- **Undergraduate Research Advisor** (Team of 2 undergraduates) *Summer 2023 – Fall 2024*
  *Pepperdine University*
- **Math Club Vice-Principal Faculty Advisor** *Spring 2023 – Fall 2023*
  *Pepperdine University*
- **Undergraduate Student Mentor** *Fall 2025, Spring 2023 – Fall 2023*
  *Pepperdine University, Faculty/Student Mentor Program*
- **Undergraduate Student Mentor** *Fall 2021 – Spring 2022*
  *University of Arizona, Women in Science and Engineering (WISE) Program*
- **Graduate Student Mentor** *Fall 2021 – Spring 2022*
  *University of Arizona, Association for Women in Mathematics (AWM) Mentor Network*
- **Undergraduate Research Advisor** *Fall 2020 – Spring 2021*
  *University of Arizona*
- **Mathematics Undergraduate Teaching Assistantship (UTA) Program Mentor** *Fall 2020 – Spring 2021*
  *University of Arizona*
- **Mathematical Modeling Group Mentor** (Team of 4 undergraduates) *Spring 2020*
  *University of Arizona*
- **Math Club Advisor** *Fall 2017 – Spring 2019*
  *The Webb Schools of California*

### University and Departmental Service

- **Creator and principal faculty advisor for Tidy Thursday** *Spring 2026, Fall 2024*
  *Pepperdine University*
  - An unofficial student club focused on exploring and analyzing datasets in `R` to improve data analysis and visualization skills.
- **Institutional Review Board, Natural Science Representative** *Fall 2023 – Present*
  *Pepperdine University*
  - Review research study protocol/application materials (exempt, expedited, and full review) and evaluate them from the perspective of the regulatory criteria for approval.
- **Admissions and Scholarship Committee** *Fall 2025 – Present, Fall 2023 – Fall 2024*
  *Pepperdine University*
  - Review student applications and select Faculty-Staff Scholars based on merit, including academic performance and recommendations.
- **Co-Instructor of RISE/Connection/Welcome Chapel** *Fall 2025 – Present, Fall 2023 – Fall 2024*
  *Pepperdine University*
  - Facilitated discussions on different aspects of science from the Christian perspective, primarily in the disciplines of history, physics, astronomy, and mathematics, with students over five sessions

- **Member of Hiring Committee, Assistant Professor of Mathematics (AY 24/25)** *Fall 2023 – Spring 2024*
  *Pepperdine University*
  - Served on the seven-member review committee whose duties included reading through application packages, conducting virtual interviews via Zoom and in-person interviews on campus, collecting and consolidating information about the interviews, and making final recommendations

- **Member of Hiring Committee, Visiting Assistant Professor of Mathematics (AY 23/24)** *Summer 2023*
  *Pepperdine University*
  - Served on the five-member review committee whose duties included reading through application packages, conducting virtual interviews via Zoom, collecting and consolidating information about the interviews, and making final recommendations

- **Member of Hiring Committee, Assistant Professor of Mathematics (AY 23/24)** *Fall 2022 – Spring 2023*
  *Pepperdine University*
  - Served on the seven-member review committee whose duties included reading through application packages, conducting virtual interviews via Zoom and in-person interviews on campus, collecting and consolidating information about the interviews, and making final recommendations

- **Member of Review Committee, Excellence in Postdoctoral Mentoring Award** *Spring 2022*
  *University of Arizona*
  - Served on the three-member review committee that determined the recipient of the 2022 Excellence in Postdoctoral Mentoring Award

- **President, Postdoctoral Governance** *Fall 2021 – Spring 2022*
  *University of Arizona*
  - Serve as an in-between for the postdocs and the Postdoctoral Committee, and organize the postdoctoral professional development seminar topics and panels

- **Mathematics Undergraduate Teaching Assistantship (UTA) Program, Director** *Fall 2021 – Spring 2022*
  *University of Arizona*
  - Coordinate the mentorship of the UTA's, and run the weekly professional development seminar

- **Mathematics Undergraduate Teaching Assistantship (UTA) Program, Co-Director** *Fall 2020 – Spring 2021*
  *University of Arizona*
  - Supported the Director of the UTA Program, and was responsible for additional duties related to the weekly professional development seminar

- **Vice President, Postdoctoral Governance** *Fall 2020 – Spring 2021*
  *University of Arizona*
  - Supported the President of the Postdoctoral Governance, and was responsible for additional duties related to the postdoctoral professional development seminars

- **Non-Academic Liaison, Postdoctoral Governance** *Spring 2020*
  *University of Arizona*
  - Organized a panel pertaining to non-academic careers for the postdoctoral professional development seminar

### Service to the Discipline

- **American Mathematical Society Spring Western Sectional Meeting 2026, Co-Organizer of Special Session** *March 2026*
  *Boise State University*
  - "Metric Graphs: Theory and Applications"

- **Webinar Panelist** *Feb 2026*
  *Transform Learning*
  - "Thinking About Thinking: Using Formative Practice to Grow Metacognitive Learners"

- **Joint Mathematics Meeting 2024, Co-Organizer of MAA Project NExT Session** *Jan 2024*
  *Pepperdine University*
  - "Making Student Thinking Visible with Team-Based Inquiry Learning"

- **MAA MathFest 2022, Co-Organizer of Themed Contributed Paper Session** *Aug 2022*
  *University of Arizona*
  - "Math Circles: Talks about Mathematical Joy, Inspirations, and Data-Driven Lessons Learned"

- **Reviewer for:**
  - Journal of Statistics and Data Science Education *Feb 2026*
  - Applied Network Science *Dec 2025*
  - Complexity *March 2023*
  - Indian Journal of Discrete Mathematics *Nov 2020*
  - DNA and Cell Biology *Jan 2020*
  - Revista de Matemática: Teoría y Aplicaciones *Oct 2019*

- **Mathematics and MATLAB Summer Workshop, Co-Coordinator** *June 2016 – June 2018*
  *Claremont Graduate University*

- **Mathematics and MATLAB Summer Workshop, Co-Instructor** *June 2016 – June 2017*
  *Claremont Graduate University*

**Outreach**

- **How I Found My Network: My Path to Mathematics** *Nov 2021*
  *Arizona State University*
  - Keynote address for Sonia Kovalevsky Day

- **Tucson Math Circle** *Aug 2019 – Present*
  *University of Arizona*
  - Co-develop materials and co-run the university sponsored weekly program designed to get middle school students excited about mathematics through hands-on exploration and discovery

- **Association for Women in Mathematics (AWM): Sonia Kovalevsky Day** *April 2021*
  *University of Arizona*
  - Developed materials and co-ran a workshop designed to bolster female high school and middle school students' passion and enthusiasm for mathematics in a supportive environment

- **The Seven Bridges of Königsberg** *Nov 2022*
  *Pepperdine University*
  - Talk given to Math Club during Tuesday Tea

- **Using Network Centrality Measures to Identify Unknown Regulatory Pathways in Pediatric Glioma** *Sept 2020*
  *University of Arizona*
  - Talk given to The MathCats Club (undergraduate math club)

# HONORS AND AWARDS

**The Teaching and Service Award** *April 2022*
*University of Arizona, Department of Mathematics*

**The Jean E. Miller Excellence in Teaching Award** *June 2018*
*The Webb Schools of California*

**The Thompson and Vivian Webb Excellence in Teaching Award** *June 2015*
*The Webb Schools of California*

**The Heinrich W. Brinkmann Mathematics Prize** *June 2012*
*Swarthmore College*

# FUNDING

**Research Grants**

- **Academic Year Undergraduate Research Initiative** ($1,000) *Sept 2025, Sept 2023*
  *Pepperdine University*
- **Collaborative Research Grant for Postdocs** ($1,500) *June 2020*
  *University of Arizona*

**Travel Awards**

- **AIM SQuaRE Research Collaboration** *July 2026, April 2025*
  *American Institute of Mathematics (AIM)*
- **TDA-BIO** ($1,000) *Oct 2016*
  *ACM Conference on Bioinformatics, Computational Biology, and Health Informatics*

**Fellowships**

- **Clinic Fellowship** ($900) *Jan 2020*
  *University of California, Irvine*
- **Daniel Pick Fellowship** ($10,000) *Oct 2017*
  *Claremont Graduate University*
- **Joseph and Elizabeth Peeler Endowed Fellowship** ($32,570) *Aug 2015 – June 2017*
  *Claremont Graduate University*
- **CGU Mathematics Fellowship** ($13,700) *Aug 2014 – June 2015, June 2017*
  *Claremont Graduate University*
- **CGU Minority Fellowship** ($2,000) *Aug 2014 – June 2016*
  *Claremont Graduate University*

# SKILLS

**Programming Languages**

- C (*Moderate proficiency*)
- C++ (*Moderate proficiency*)
- MATLAB (*Proficient*)
- Python (*Proficient*)
- R (*Proficient*)

**Scientific Applications**

- GitHub
- LaTex
- RSweave